

Industrial Internship Report on

“PYTHON-BASED QUIZ GAME AND PERFORMANCE ANALYSIS SYSTEM”

Prepared by

Priyanshi Choudhary

Python Learning Internship

Organized by

UPSKILL CAMPUS

The IoT ACADEMY

&

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Executive Summary

The **Python-Based Quiz Game and Performance Analysis System** was developed as part of a four-week Python internship program. The project is a console-based application that allows users to take topic-based quizzes on Python, Pandas and NumPy.

The system loads questions from a CSV file, validates user answers, calculates scores and percentages, and classifies overall performance. Pandas is used for data handling and storing quiz results, NumPy is used for statistical analysis, and Matplotlib is used to visualize correct and incorrect answers.

This project provided practical experience in Python programming, CSV file handling, data analysis, visualization, input validation and developing a complete menu-driven application.

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1. Preface

Industrial internships provide students with an opportunity to understand how the concepts learned in an academic environment can be applied to practical software development. They help bridge the gap between theoretical knowledge and real-world implementation by providing hands-on experience with programming languages, libraries, development tools and problem-solving techniques.

As part of the four-week Python internship program, this project was developed to apply fundamental Python programming concepts in a practical application. The selected project, Python-Based Quiz Game and Performance Analysis System, combines a simple quiz application with data analysis and visualization features.

The project was designed to allow users to select a topic, answer multiple-choice questions and receive an automatically calculated result. The system also maintains quiz results so that previous attempts can be analyzed. The project includes three topics: Python, Pandas and NumPy, with ten questions available for each topic.

During the development process, concepts such as conditional statements, loops, functions, input validation, CSV file handling and data processing were applied. Additional libraries such as Pandas, NumPy and Matplotlib were used to demonstrate practical data handling, numerical analysis and visualization.

The internship also provided exposure to project organization and version control using GitHub. The completed project and its supporting files were maintained in a public GitHub repository.

2. Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is a technology company established in 2013 that works in the field of digital transformation and provides technology-based industrial solutions.

The company works with technologies such as Internet of Things (IoT), Cloud Computing, Machine Learning, Cyber Security, Python, Java, communication technologies, and front-end development.

UCT develops solutions for industrial and business applications, including IoT platforms, smart factory solutions, predictive maintenance, and data-driven monitoring systems.

The organization also focuses on applying modern technologies to solve practical industrial problems and improve operational efficiency.

During the internship program, UCT served as the industrial partner and provided the overall learning and project framework through the internship program.

2.2 About Upskill Campus

Upskill Campus is a career development and learning platform focused on helping students and learners develop industry-relevant technical and professional skills.

As part of the internship program, Upskill Campus facilitated the learning process and provided access to learning resources, project activities, and internship-related guidance.

The Python internship provided structured learning material covering fundamental Python programming concepts and introduced additional libraries such as NumPy and Pandas.

The internship also encouraged practical application of the concepts learned through the development of a Python-based Student Management System.

2.3 About The IoT Academy

The IoT Academy is an educational initiative associated with UniConverge Technologies Pvt. Ltd. that provides technical training and certification programs in various technology domains.

It focuses on providing learners with industry-oriented knowledge and practical exposure through structured training programs.

As part of this internship, The IoT Academy was involved in facilitating the learning and internship process along with Upskill Campus and UniConverge Technologies Pvt. Ltd.

2.4 Objectives of the Internship

The major objectives of the four-week Python internship were:

- To develop a basic understanding of Python programming.
 - To understand conditional statements, loops, functions, and data structures.
 - To learn basic exception handling and file handling.
 - To gain an introduction to NumPy and Pandas.
 - To apply Python programming concepts to a practical project.
 - To develop problem-solving and logical thinking skills.
 - To gain experience in organizing and maintaining source code.
 - To understand basic Git and GitHub version-control practices.
 - To gain practical exposure to software development through project implementation.
-

2.5 References

1. Python Documentation — Python programming language reference and documentation.
 2. NumPy Documentation — Reference for numerical operations and statistical functions.
 3. Pandas Documentation — Reference for DataFrame and data-processing operations.
 4. Git Documentation — Reference for version control and Git commands.
 5. GitHub Documentation — Reference for repository and source-code management.
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2.6 Glossary

Term	Description
Python:	A high-level programming language used to develop the project.
CSV:	Comma-Separated Values, a file format used for storing tabular data.
NumPy:	A Python library used for numerical and mathematical operations.
Pandas:	A Python library used for data manipulation and analysis.
Quiz:	A set of questions used to evaluate a user's knowledge.
Percentage:	The percentage of questions answered correctly out of the total number of questions.
Performance Analysis:	The process of evaluating quiz results to understand performance.
Git	A distributed version control system used to track changes in source code.
GitHub	An online platform used to host and manage Git repositories.
IDE	Integrated Development Environment used for writing and managing code.
DataFrame:	A two-dimensional Pandas data structure used to store and process tabular data.
Data Visualization:	The representation of data using graphical elements such as charts.

3. Problem Statement

Traditional quizzes are often conducted manually, where questions are presented to users and answers are checked manually. This approach requires additional effort for calculating scores, determining percentages and maintaining records of previous attempts.

Another limitation of a basic quiz system is the lack of performance analysis. Simply knowing the final score does not provide enough information to understand overall performance. When multiple attempts are made, manually comparing the results can also become difficult.

The objective of this project is to develop a simple automated quiz application that can conduct topic-based quizzes and calculate results without requiring manual calculations. The application should provide multiple-choice questions, accept user responses, validate the input and automatically calculate the final result.

The project also aims to provide basic performance analysis. Quiz results should be stored so that previous attempts can be examined. Numerical analysis can be performed using NumPy, while graphical visualization can help users understand their performance more easily.

Therefore, the proposed system combines quiz management, automatic evaluation, result storage and performance analysis into a single Python application.

4. Existing and Proposed Solution

4.1 Existing Solution

In a traditional manual quiz process, questions may be stored or prepared separately and presented to students manually. After the quiz, the answers need to be checked and the score has to be calculated.

Such an approach has several limitations:

- Manual checking requires additional time.
- Score calculation can require repetitive work.
- Previous results may not be organized properly.
- Comparing multiple attempts can be difficult.
- Performance statistics may need to be calculated manually.
- Graphical representation of results is generally unavailable.
- Invalid inputs may not be handled consistently.

A basic computerized quiz can solve some of these problems, but without proper data handling and analysis, it may still provide only limited functionality.

4.2 Proposed Solution

The proposed system automates the complete basic quiz process.

Questions are stored in a CSV file and loaded into the application using Pandas. The user selects a topic and answers the questions displayed by the program. The application checks the user's response against the correct answer and keeps track of the score.

After the quiz is completed, the system calculates the number of correct and incorrect answers and determines the percentage. It also classifies the user's performance.

The results are stored so that they can be used for future analysis. NumPy is used for statistical calculations, and Matplotlib provides a graphical representation of quiz performance. The project therefore combines programming, data handling, numerical analysis and visualization in one application.

4.3 Value Addition

Topic-Based Quiz

The application allows users to select from three available topics: Python, Pandas and NumPy. There are 30 questions in total, with 10 questions for each topic.

Multiple-Choice Questions

Each quiz contains multiple-choice questions with answer options. The user selects an answer and the system checks it automatically.

Answer Validation

The application validates the user's input so that inappropriate options can be rejected and the user can provide a valid answer.

Automatic Score Calculation

The system automatically increases the score when the user selects the correct answer and calculates the final score after the quiz.

Percentage Calculation

The final percentage is calculated based on the number of correct answers and the total number of questions.

Performance Classification

The system categorizes the user's performance based on the obtained result, helping the user understand the quality of their performance.

Result Storage

Quiz results can be stored for later use and analysis. Pandas is used for handling the result data.

Statistical Analysis

NumPy is used for numerical analysis, including calculations such as average, maximum and minimum values.

Graphical Visualization

Matplotlib is used to generate a graph showing the relationship between correct and incorrect answers.

Overall Performance Analysis

The system provides an option to analyze previous quiz attempts and obtain an overall view of performance.

4.4 Code Submission

The complete source code of the Student Management System has been uploaded to the GitHub repository created for the internship.

GitHub Code Link:

<https://github.com/priyanshi25scs1003002331-iilm/upskillcampus/blob/main/QuizGame.py>

4.5 Report Submission

The final internship report will be uploaded to the same GitHub repository along with the project source code.

GitHub Report Link:

https://github.com/priyanshi25scs1003002331-iilm/upskillcampus/blob/main/QuizGame_Performance_Analysis_Report_Priyanshi_USC_UCT.pdf

5. Proposed Design / Model

5.1 High Level Diagram

The user first interacts with the main menu and chooses whether to start a quiz or view overall performance. When a quiz is started, the application loads questions from the CSV dataset and presents them to the user.

After the user submits answers, the application processes the responses and calculates the result. The result is then passed to the analysis components, where NumPy can perform numerical calculations and Matplotlib can create a visual representation.

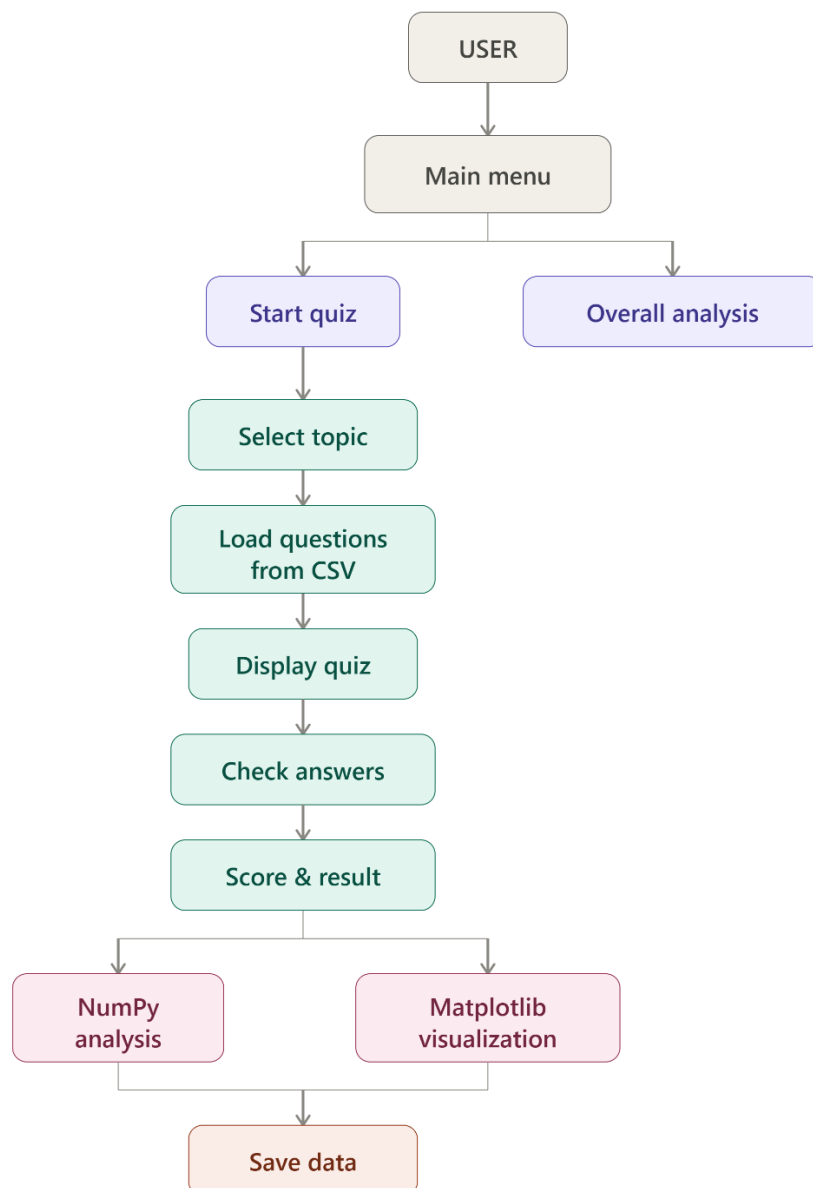
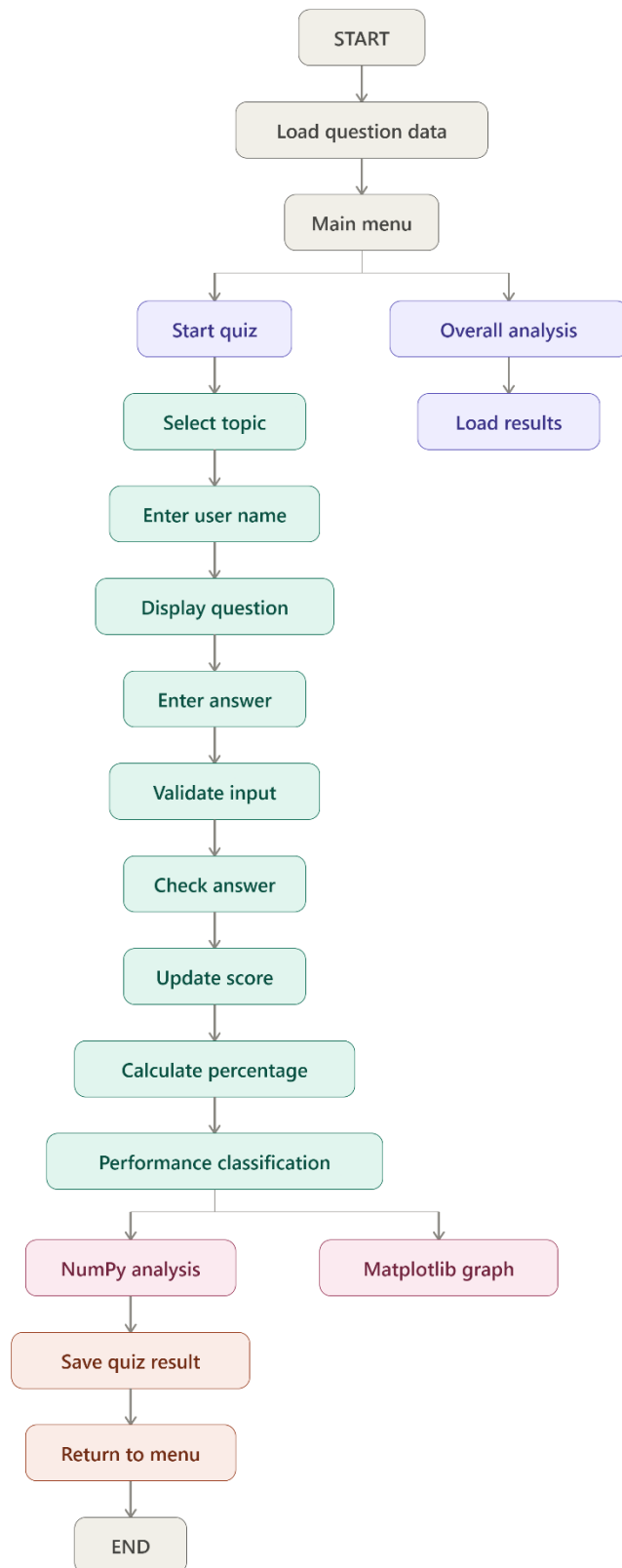


Figure 1 : High-Level Architecture of the Quiz Game and Performance Analysis System

5.2 Low Level Diagram



When the program starts, the user is presented with the main menu. After selecting the quiz option, the user chooses a topic and provides the required information. Questions are then displayed one by one.

The system validates each answer, compares it with the correct answer and updates the score. At the end of the quiz, the final result is calculated, classified and stored.

Figure 2 : Low-Level Functional Flow of the Student Management System

5.3 Interfaces

The application uses a simple console-based interface that allows users to interact with the system through keyboard input.

Main Menu Interface

The main menu acts as the starting point of the application. It allows the user to select the required operation, such as starting a quiz or viewing overall performance.

```
=====
                        QUIZ GAME & ANALYZER
=====
1. Start Quiz
2. View Overall Performance
3. Exit
=====
Enter your choice: |
```

Figure 3: Quiz Main Menu

Topic Selection Interface

The topic selection section allows the user to choose a subject from the available topics. The current project provides Python, Pandas and NumPy as quiz categories.

```
Available Topics:
1. Python
2. Pandas
3. NumPy

Select a topic:
```

Figure 4: Quiz Topic Selection

Quiz Interface

The quiz interface displays the question and available multiple-choice answers. The user enters the selected option through the keyboard.

```
=====
                        PYTHON QUIZ GAME
=====
Enter your name: Priyanshi

Welcome, Priyanshi
Topic: Python
Let's start the quiz!

Question 1
Which keyword is used to define a function in Python?
A. func
B. define
C. def
D. function
Your answer:
```

Figure 5: Interface of the Quiz

Result Interface

After completing the quiz, the result section displays the user's performance, including the score, percentage and performance classification.

```
=====
                        QUIZ RESULT
=====
Student Name : Priyanshi
Topic : Python
Total Questions : 10
Correct Answers : 6
Wrong Answers : 4
Score : 6 / 10
Percentage : 60.0 %
Average Answers : 5.0
Maximum Answers : 6
Minimum Answers : 4
Performance : Good
=====

Result saved successfully to results.csv
```

Figure 6: Quiz Result Showing

Performance Analysis Interface

The performance analysis section uses stored results to provide statistical information about previous quiz attempts.

```
=====
                        OVERALL PERFORMANCE ANALYSIS
=====
Total Attempts          : 3
Average Score           : 5.0
Highest Score           : 6
Lowest Score            : 4
Average Percentage      : 50.0 %
Best Performing Student : Priyanshi
=====
```

Figure 7: Quiz Performance Analysis

Visualization Interface

Matplotlib generates a graphical representation of correct and incorrect answers, providing a visual way to understand quiz performance.

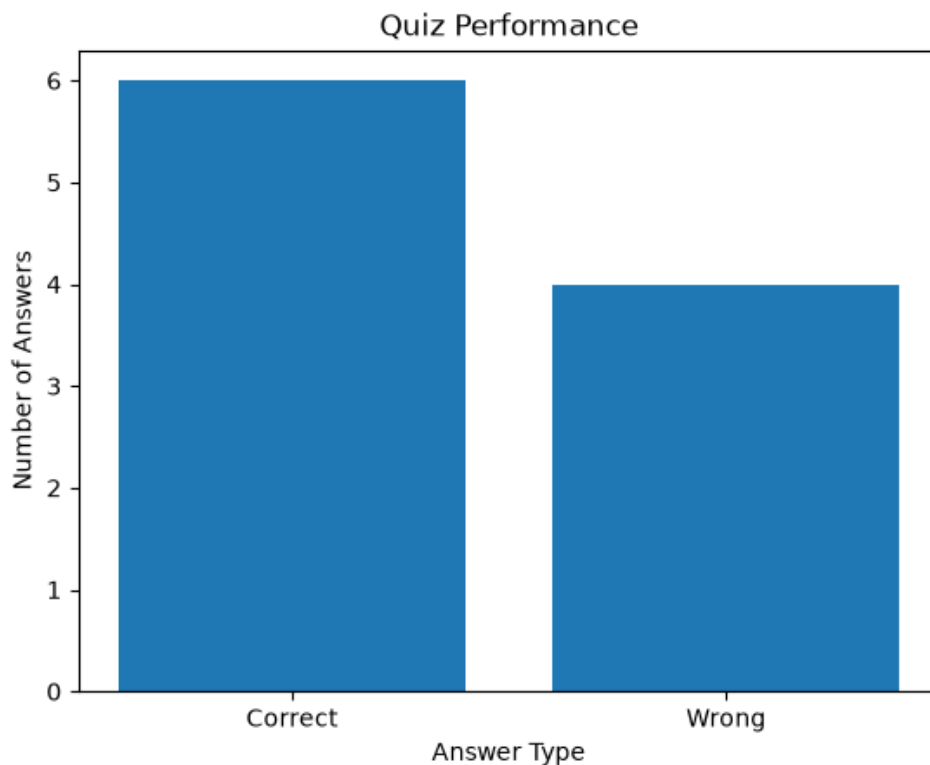


Figure 8: Diagram of the Result

6. Performance Test

6.1 Test Plan / Test Cases

The following test cases were used to verify the major functionalities of the system.

Test Case ID	Test Case	Steps / Input	Expected Result	Actual Result	Status
TC01	Program Launch	Run QuizGame.py	Program starts successfully and displays the main menu	Main menu displayed successfully	Pass
TC02	Main Menu Selection	Select option 1 – Start Quiz	Topic selection menu is displayed	Topic selection displayed	Pass
TC03	Topic Selection	Select Python topic	Python quiz questions are loaded	Python questions loaded successfully	Pass
TC04	Topic Selection	Select Pandas topic	Pandas quiz questions are loaded	Pandas questions loaded successfully	Pass
TC05	Topic Selection	Select NumPy topic	NumPy quiz questions are loaded	NumPy questions loaded successfully	Pass
TC06	Invalid Topic Input	Enter an invalid topic number	System displays an error and asks for valid input	Error displayed and valid input requested	Pass
TC07	User Name Input	Enter a valid student name	Name is accepted and quiz starts	Name accepted successfully	Pass
TC08	Question Display	Start a quiz	Question and four answer options (A-D) are displayed	Question and options displayed	Pass
TC09	Valid Answer Input	Enter A, B, C or D	Answer is accepted and next question is displayed	Answer accepted successfully	Pass

TC10	Invalid Answer Input	Enter an invalid answer such as X	System asks the user to enter A, B, C or D	Invalid input handled correctly	Pass
TC11	Answer Validation	Enter a correct answer	Correct answer counter is increased	Score updated correctly	Pass
TC12	Answer Validation	Enter an incorrect answer	Wrong answer counter is increased	Wrong answer counted correctly	Pass
TC13	Score Calculation	Complete all quiz questions	Final score is calculated correctly	Score calculated correctly	Pass
TC14	Percentage Calculation	Complete the quiz	Percentage is calculated from the score	Percentage displayed correctly	Pass
TC15	Performance Classification	Complete quiz with different scores	Performance is classified as Excellent, Good, Average or Needs Improvement	Classification displayed correctly	Pass
TC16	Result Storage	Complete a quiz	Result is stored in results.csv using Pandas	Result stored successfully	Pass
TC17	Overall Performance	Select option 2 from main menu	Overall attempts, average, highest and lowest scores are displayed	Performance analysis displayed correctly	Pass
TC18	Statistical Analysis	Complete a quiz	NumPy calculates statistical values such as mean, maximum and minimum	Statistics calculated successfully	Pass
TC19	Graphical Visualization	Complete a quiz	Bar graph showing correct and wrong answers is displayed	Graph displayed successfully	Pass
TC20	Exit Program	Select option 3 – Exit	Program terminates successfully	Program exited successfully	Pass

6.2 Test Procedure

The application was tested systematically to verify its functionality.

First, the Python program was executed and the main menu was checked. The available menu options were then tested to ensure that the program responded correctly.

Each available quiz topic was selected individually. Questions were displayed and valid answers were entered to verify the normal quiz flow. Incorrect answers were also entered to confirm that they were recorded properly.

Invalid inputs were tested to verify that the application did not accept unsupported answer choices. This helped confirm the input validation mechanism.

After completing a quiz, the score and percentage were checked. The performance classification was also verified based on the obtained result.

The result storage functionality was then checked to confirm that quiz information could be maintained for later analysis.

Finally, the overall performance option and graphical visualization were tested to verify that stored data could be analyzed and represented correctly.

6.3 Performance Outcome

The testing process showed that the application successfully performs the primary operations expected from a quiz and performance analysis system.

The system can load quiz questions, allow users to select a topic, accept answers and calculate the final score. The project supports three topics and contains 30 questions in total.

The application also handles incorrect inputs and provides appropriate validation. After completing a quiz, the system calculates the percentage and classifies the user's performance.

The use of Pandas allows the project to work with structured quiz and result data. NumPy provides numerical analysis capabilities, while Matplotlib adds graphical visualization.

Overall, the testing demonstrated that the developed system meets its primary requirements and provides a functional platform for conducting quizzes and analyzing performance.

7. My Learnings

The development of this project provided practical experience in several areas of Python programming and software development.

Python Programming

I improved my understanding of Python syntax, variables, conditional statements, loops and functions. These concepts were used together to create the different operations of the quiz application.

Functions and Program Structure

The project helped me understand how a larger Python program can be divided into functions. This makes the code easier to understand, maintain and modify.

Input Validation

I learned the importance of validating user input. A user may enter an unexpected value, so the program should handle such situations without crashing.

CSV File Handling

The project provided practical experience in working with CSV files. Quiz questions and results can be stored in a structured format and accessed by the program when required.

Pandas

I learned how Pandas can be used to read and process tabular data. It was particularly useful for handling the quiz dataset and result information.

NumPy

I learned how NumPy can simplify numerical calculations and statistical operations. It was used to analyze quiz performance data.

Matplotlib

The project introduced the basic use of Matplotlib for data visualization. Instead of displaying performance only as numbers, the application can represent results graphically.

Data Analysis

The project helped me understand the basic process of converting raw data into useful information. Quiz results can be processed to calculate averages, maximum and minimum values and other performance-related information.

Testing and Debugging

Testing the project with different valid and invalid inputs helped me understand how errors can occur and how they can be identified and corrected.

Git and GitHub

Maintaining the project on GitHub provided practical experience with version control, repository management and project documentation.

Overall, this project helped me understand how different programming concepts and libraries can be combined to develop a complete working application rather than treating each concept as an isolated programming exercise.

8. Future Work Scope

Although the current application provides the main functionality required for conducting quizzes and analyzing results, several improvements can be introduced in future versions.

More Quiz Topics

Additional topics can be added to make the system useful for a wider range of subjects. Topics could include C++, Java, HTML, CSS, JavaScript, databases and computer networks.

Larger Question Database

The number of questions can be increased so that users receive a greater variety of questions during different attempts.

Random Question Selection

Questions can be randomized so that every quiz attempt does not follow the same order. This can make the quiz more engaging and reduce the possibility of memorizing the question sequence.

Difficulty Levels

Questions can be divided into Easy, Medium and Hard categories. Users could then select their preferred difficulty level.

Timer

A timer could be introduced to make the quiz more realistic and test the user's ability to answer questions within a limited period.

Graphical User Interface

The current console-based application can be converted into a graphical application using Tkinter or another Python GUI framework. This would make the system more user-friendly.

Database Integration

Instead of storing results only in CSV files, a database such as SQLite or MySQL could be used. This would make the application more suitable for handling a larger number of users and records.

User Authentication

Login and registration features could be added to allow individual users to maintain their own quiz history.

Detailed Performance Dashboard

A dashboard could display topic-wise scores, historical performance, average scores and improvement over time.

Administrator Panel

An administrator interface could be developed to add, edit and delete quiz questions without directly modifying the CSV file.

Advanced Data Analysis

Additional statistical techniques could be implemented to identify strong and weak topics, compare users and track performance trends.

Online Quiz System

The project could eventually be converted into a web-based application where users can access quizzes through a browser and results can be stored centrally.

These improvements could transform the current educational quiz application into a more complete assessment and performance-monitoring platform.